

SURGICAL, CHEAP, AND FLEXIBLE: MITIGATING FALSE REFUSAL IN LANGUAGE MODELS VIA SINGLE VECTOR ABLATION

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ABSTRACT

Training a language model to be *both* helpful and harmless requires careful calibration of refusal behaviours: Models should refuse to follow malicious instructions or give harmful advice (e.g. “how do I kill someone?”), but they should *not* refuse safe requests, even if they superficially resemble unsafe ones (e.g. “how do I kill a Python process?”). Avoiding such *false refusal*, as prior work has shown, is challenging even for highly-capable language models. In this paper, we propose a simple and surgical method for mitigating false refusal in language models via single vector ablation. For a given model, we extract a false refusal vector and show that ablating this vector reduces false refusal rate while preserving the model’s safety and general capabilities. We also show that our approach can be used for fine-grained calibration of model safety. Our approach is training-free and model-agnostic, making it useful for mitigating the problem of false refusal in current and future language models.¹

1 INTRODUCTION

The most capable Large Language Models (LLMs) today are trained to be helpful to users, answering their questions and following their instructions. However, LLMs trained *only* to be helpful will follow even malicious instructions and readily generate harmful content (Bianchi et al., 2023). Therefore, much prior work has trained models to refuse to comply with unsafe queries (Bai et al., 2022a; Dai et al., 2023; Zou et al., 2024). This creates a tension between model ‘helpfulness’ and ‘harmlessness’, and thus requires careful calibration, which is difficult to achieve: Röttger et al. (2024) show that even highly capable LLMs struggle with *false refusal*, where they refuse to comply with clearly safe queries just because they superficially resemble unsafe queries (e.g. “how do I make someone *explode* with laughter?”). This makes LLMs less helpful.

Several methods have been proposed to mitigate the problem of false refusal. However, all have clear limitations. Training-based methods, on the one hand, may be effective in reducing false refusal, but they are also *inflexible*, since safety can only be calibrated at training time (Zhang et al., 2024; Zheng et al., 2024). Training-free methods, on the other hand, provide more flexibility but are *costly*, because they require expensive computation at inference time (Cao et al., 2024; Shi et al., 2024). Furthermore, training-free methods so far appear *imprecise*, because they have unintended negative effects on general model capability (Cao et al., 2024).

In this paper, we propose a new method for mitigating false refusal in LLMs via single vector ablation, which addresses these key limitations of prior work. We extract **true refusal vector** and **false refusal vector** with a small sample of harmful and pseudo-harmful queries, and apply orthogonalization to tease the two vectors apart. Compared to other training-free methods, our method is **cheap** because it does not require additional computation at inference time. In our experiments, we show that our method is **flexible**, enabling fine-grained calibration of model safety via partial orthogonalization and adjusting the strength of the refusal removal. We also show that our method is **surgical**, reducing false refusal rates while keeping the safety and general capability of the model.

¹We release our code at <https://github.com/mainlp/False-Refusal-Mitigation>.

2 BACKGROUND

2.1 REFUSAL VECTOR EXTRACTION

In prior work, Zou et al. (2023a) and Arditi et al. (2024) used *difference-in-means* (Belrose, 2023) to extract a refusal vector from model activations. They focused on general refusal behaviour, regardless of whether it was true or false refusal. The candidate refusal vectors were extracted by calculating the difference between the mean activations when prompted with harmful queries $\mathcal{D}_{\text{harmful}}$ and those when prompted with harmless queries $\mathcal{D}_{\text{harmless}}$ at layer l and token position i :

$$\mathbf{r}_{i,l} = \mathbf{v}_{i,l}^{\text{harmful}} - \mathbf{v}_{i,l}^{\text{harmless}} \quad (1)$$

$$\mathbf{v}_{i,l}^{\text{harmful}} = \frac{1}{|\mathcal{D}_{\text{harmful}}^{(\text{train})}|} \sum_{\mathbf{t} \in \mathcal{D}_{\text{harmful}}^{(\text{train})}} \mathbf{x}_{i,l}(\mathbf{t}), \quad \mathbf{v}_{i,l}^{\text{harmless}} = \frac{1}{|\mathcal{D}_{\text{harmless}}^{(\text{train})}|} \sum_{\mathbf{t} \in \mathcal{D}_{\text{harmless}}^{(\text{train})}} \mathbf{x}_{i,l}(\mathbf{t}) \quad (2)$$

where $\mathbf{x}_{i,l}(\mathbf{t})$ is the residual stream activation of the transformer at token position i and layer l when prompted with text t . The *diff-in-means* vectors across all layers at *post instruction* token positions, such as the `[/INST]` for Llama2 models, are collected as candidates for the final refusal vector. From all the candidate vectors, we find the vector that is the most effective in removing the refusal behaviour, by ranking them based on a drop in refusal score (as given in equation 3) after ablating it from the model’s activation stream. The refusal score measures the token probability difference between the refusal-related tokens $\mathcal{R}$ such as ‘Sorry’, ‘I’, and the non-refusal-related tokens $\mathcal{V} \setminus \mathcal{R}$, at the first token position in the model’s response:

$$\text{Refusal Score} = \log \left(\sum_{t \in \mathcal{R}} p_t \right) - \log \left(\sum_{t \in \mathcal{V} \setminus \mathcal{R}} p_t \right) \quad (3)$$

A similar approach can also be applied to find the vector that is most effective for increasing the refusal rate when added to model activations.

2.2 VECTOR ABLATION AND ADDITION

The selected refusal vector $\hat{\mathbf{r}}$ can be used to remove refusal behaviour by ablating it from the residual stream. This is done by first projecting the residual stream activation onto the direction of the refusal vector and then removing this projection from the activation:

$$\mathbf{x}' \leftarrow \mathbf{x} - \hat{\mathbf{r}} \hat{\mathbf{r}}^\top \mathbf{x} \quad (4)$$

This operation is done across all layers and token positions to remove the refusal behaviour from the model effectively. The model can be made to refuse more by adding the refusal vector to the activations at all token positions of a certain layer l :

$$\mathbf{x}'_l \leftarrow \mathbf{x}_l + \alpha \hat{\mathbf{r}}_l \quad (5)$$

where $\hat{\mathbf{r}}_l$ is selected and applied to the activation $\mathbf{x}_l$ from the same layer. $\alpha \in [0, 1]$ controls the strength of the addition operation. Arditi et al. (2024) show that it is sufficient to do vector addition at a single layer to make the model to refuse more in contrast to the removal operation where the ablation is applied at all layer activations. Therefore, we follow the same procedure in our vector ablation and addition experiments. Both operations can be directly applied to the model weights, without causing additional computation at inference time. We refer readers to Appendix D for a detailed explanation on the equivalence between inference time steering and model weights editing.

3 MITIGATING FALSE REFUSAL VIA SINGLE VECTOR ABLATION

We aim to reduce unwanted false refusal behaviour of language models with minimal effect on true refusal behaviour and performance on general tasks. This requires us to find false refusal related features that have minimal overlap with those associated with true refusal and general tasks. We first show that simply using the *difference-in-means* technique cannot successfully find false refusal features with minimal overlap with other features. To disentangle the features, our key suggestion is to orthogonalize the false refusal and true refusal vectors to avoid harming the true refusal ability when ablating the orthogonalized false refusal vector.

3.1 EXTRACTING A FALSE REFUSAL VECTOR

Similar to extracting the true refusal vector in equation 1, we extract the refusal vector for false refusal prompts by replacing the harmful queries $\mathcal{D}_{\text{harmful}}$ with pseudo-harmful queries $\mathcal{D}_{\text{pseudo-harmful}}$ (e.g. “how do I kill a Python process?”), which induce false refusal in the model:

$$\mathbf{w}_{i,l} = \mathbf{v}_{i,l}^{\text{pseudo-harmful}} - \mathbf{v}_{i,l}^{\text{harmless}} \quad (6)$$

$$\mathbf{v}_{i,l}^{\text{pseudo-harmful}} = \frac{1}{|\mathcal{D}_{\text{pseudo-harmful}}^{(\text{train})}|} \sum_{\mathbf{t} \in \mathcal{D}_{\text{harmful}}^{(\text{train})}} \mathbf{x}_{i,l}(\mathbf{t}), \quad (7)$$

We use the refusal score as a filter to select samples that have a refusal score larger than zero. We collected 128 pseudo-harmful samples for extracting the vector which was shown to be effective in Ardit et al. (2024). Then, we select the most effective vector $\hat{\mathbf{w}}$ by validating the refusal score (introduced in Eq. 3) drop on the validation set consisting of 32 pseudo-harmful samples. More detailed dataset usage for vector construction is described in Section 4. However, as shown in Table 1, after ablating the false refusal vector from the activation stream, we see an increase in compliance rate for *both* the harmful and harmless data, similar to ablating the true refusal vector $\hat{\mathbf{f}}$. In our preliminary experiments, this issue was validated with different pseudo-harmful datasets such as OR-Bench-Hard (Cui et al., 2024b), XSTest (Röttger et al., 2024) and OKTest (Shi et al., 2024). This means that ablating the raw *diff-in-means* vector is insufficient, and most importantly, the true refusal vector $\hat{\mathbf{v}}$ and false refusal vector $\hat{\mathbf{w}}$ are not independent of each other. To tease the two vectors apart, we propose to apply orthogonalization between the candidate false refusal vectors $\mathbf{w}_{i,l}$ and the candidate true refusal vector $\mathbf{v}_{i,j}$, resulting in orthogonalized false refusal vector $\mathbf{w}'_{i,j}$. Then, we select the most effective one $\hat{\mathbf{w}}'$ as we described in the procedure above. As shown in Table 1, after ablating $\hat{\mathbf{w}}'$, the model maintains its low compliance rate on harmful data, while the compliance on harmless data increases substantially compared to the original model, showing the effectiveness of the orthogonalization operation for teasing the vectors apart.

Operation		Harmful CR ↓	ORBench-H CR ↑	XSTest-S CR ↑
LLAMA2-7B-CHAT	-	2.3	14.8	13.6
- $\hat{\mathbf{f}}$	ablating true refusal vector	93.0	100	93.0
- $\hat{\mathbf{w}}$	ablating false refusal vector	46.1	100	90.0
- $\hat{\mathbf{w}}'$	ablating orthogonalized false refusal vector	3.1	65.6	57.6

Table 1: Extracting a *diff-in-means* refusal vector is not enough: Ablating (-) any of the two *diff-in-means* vectors $\hat{\mathbf{f}}$ or $\hat{\mathbf{w}}$ will remove the refusal behaviour, regardless of whether a harmful or harmless prompt is given. Instead, ablating orthogonalized false refusal vector $\hat{\mathbf{w}}'$ successfully keeps the model’s low compliance rate on harmful data, while the compliance on harmless data increases substantially compared to the original model. CR: Compliance Rate.

3.2 PARTIAL ORTHOGONALIZATION

The orthogonalization operation so far has the benefit to disentangle the true refusal and false refusal vectors. It essentially sets a boundary to ensure any feature in the prompts related to harmful data (used for extracting the refusal vector) should trigger a refusal response. However, this will make the model overly cautious when it comes to ambiguous examples, such as a request like “*how to cut off the head of a fish*”, which a cautious model will warn against as it “*can be harmful to the fish’s health and affect the flavour*”, while a less cautious model will comply with. To address this, we show that the refusal boundary can be adjusted by partial orthogonalization, where we introduce a coefficient λ into the subtraction operation:

$$\mathbf{w}'_{i,l} \leftarrow \mathbf{w}_{i,l} - \lambda \mathbf{v}_{i,l} \mathbf{v}_{i,l}^\top \mathbf{w}_{i,l} \quad (8)$$

Partial orthogonalization allows us to flexibly adjust the refusal level of the model. By lowering the λ coefficient, the false refusal vector is modified less, so its ablation will have a stronger impact on false refusal behaviour of the model. This enables us to granularly control the model’s sensitivity to ambiguous samples, which we will discuss in Section 5.2.